

Project overview

A community early-warning and support system for power and water outages

Amethyst & Azure

Kasi Grid: Know before it goes dark.

FNB App of the Year Hackathon 2026 · Track: Climate resilience and essential services

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Source code: https://github.com/Unsolvedstar/Kasi-Grid

1. What inspired this project

Loadshedding is over in South Africa, but the outages are not. Eskom has reported more than 500 days without it, yet in the townships the power still cuts out because of load reduction, cable theft and substation faults. In Johannesburg the water fails too: 22 major outages in two weeks of January 2026 alone. When this happens there is no stage or schedule to check. People turn to rumours in WhatsApp groups, while families and spaza shops lose food, stock and income waiting to learn what is real.

We are Tshedza Tshikovhi and Mulanga Munaka, both 22. Tshedza is a software developer who leads a small team at Shudu Connections in The Innovation Hub, building an SME incubation platform, and who built CashCram, a Top-13 finalist in NWU's 2026 Pitch Your Business Competition. Mulanga is a biomedical engineering student at Wits, and her training in designing for real people in high-stakes settings shaped the medical side of the app.

Our idea is simple: the people living in a neighbourhood already know what is failing in it. Kasi Grid turns their reports into verified, community-level alerts. An outage counts as verified only when three different devices agree.

2. Problem statement

Loadshedding has officially ended. Eskom reports more than 500 days without it. But in townships and informal settlements, power and water still fail without warning. The causes are load reduction, cable theft and substation faults, and in Johannesburg 22 major water outages hit in just two weeks of January 2026. These failures have no stage and no published schedule, so there is nothing to check. People rely on rumours in WhatsApp groups to learn what is real. The people hurt most are low-income households and spaza shop owners, who lose food, stock and a day's income because they find out too late to prepare. Households that depend on power for oxygen concentrators or other medical devices are at the greatest risk, because they cannot tell how long their battery will last against an outage.

The pain points are specific. There is no trusted, community-level source of what is failing right now. There is no warning before it happens. Rumour spreads faster than any official notice. And the tools that exist assume a signal and a schedule, but signal drops exactly when the power does. The people living in a neighbourhood already know what is failing in it, but nothing turns that knowledge into something others can act on.

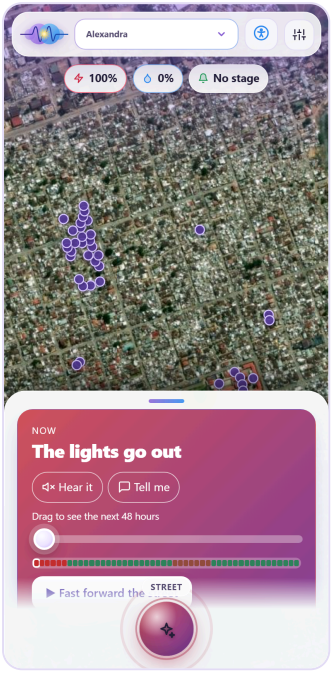
Sources: Eskom statements and SAnews (October 2026); Daily Maverick (January 2026); allAfrica (August 2026).

3. Solution and features

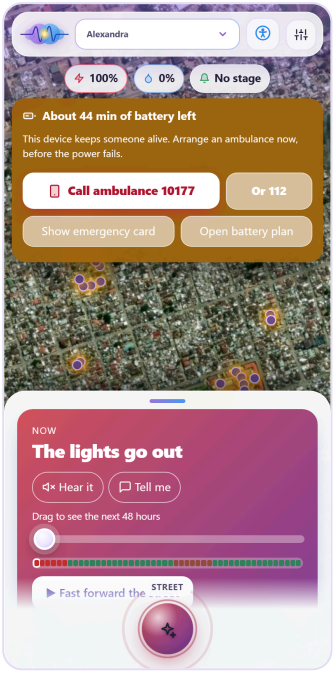
Kasi Grid is a free web app that works anonymously (an account is optional), installs on a phone and keeps working offline. It turns neighbours' reports into verified outage alerts for power and water, learns when and how long outages happen, and connects an outage to action: a medical battery countdown, an ambulance call, help from neighbours and what an outage costs a shop. It works for any place in South Africa.

- **A live satellite street.** The home screen is a real satellite view of the community with every mapped home as a glowing dot. Homes go dark as the risk of a power cut rises. Drag a slider through the next 48 hours and watch the street dim and relight. Tap a home for its status.
- **Verified outages.** Three or more different devices reporting the same problem make a verified outage on the map. Neighbours vote "back on for me" and the alert closes itself. Reports are anonymous.

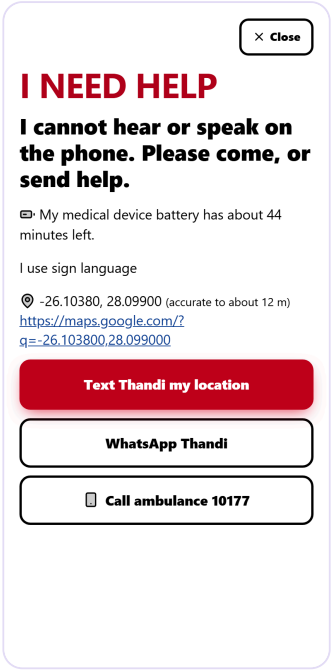
- **No account needed, and one if you want it.** Everything works anonymously. An optional free account (cell number or email and a password) keeps your area, language, accessibility settings and bookings on any phone, and signs you in automatically every time the app opens.
- **Anywhere in South Africa.** Every place is a community of about 2 km. Find any town, suburb or street, or use your location. Reports and posts can carry an exact pin; neighbours see board posts only to about 100 m so no home is given away.
- **Early warning.** The app learns each community's own pattern (which weekday and time outages cluster) and how long they last, shows "expected back around 16:30", and says when it does not have enough data.
- **Medical power plan.** A household enters its device and battery; the plan counts the real hours, tracks them live against the time on battery (and against readings from the battery display), alerts at 2 hours, 1 hour, 30 minutes and empty, and puts a call to the ambulance one tap away. A text-my-location emergency card works without a voice call.
- **Care and equipment booking** in four short steps: fixed price, the nearest carers or runners are asked, a prescription check for oxygen.
- **Small business outage cost.** What an outage has cost a shop so far in lost sales and stock at risk, and ranked fixes by payback.
- **Help between neighbours.** A community board for offers and requests (charging, water, lifts, cold storage), restock orders pooled between spazas, nearby shops, clinics and food prices, savings calculators.
- **Built for everyone.** Accessibility settings for people who are deaf, hard of hearing, blind or have low vision (flash, vibration, large-text and spoken alerts, screen-reader announcements, text size, high contrast, reduced motion). Four languages, offline, low data.



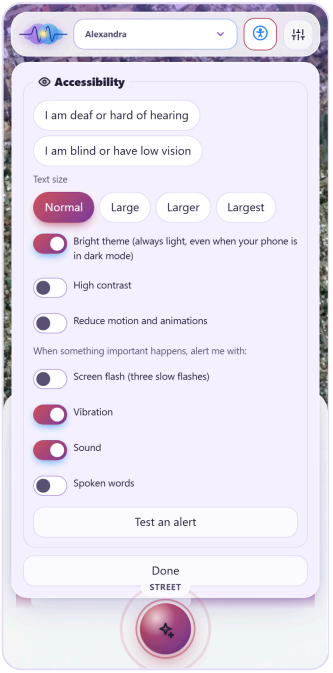
The live satellite street



Battery countdown and ambulance



Emergency card, no call needed



Accessibility settings

4. What makes this solution stand out

Most outage apps are built for loadshedding: they assume Eskom publishes a stage and a schedule, and they display it. That no longer fits what townships live with. Kasi Grid is built for the failures that have no schedule: load reduction, cable theft, substation faults and water cuts. It gets its information from the one source that is always there, the neighbours sitting in the dark.

- **Trustworthy.** One report is not treated as truth. An outage is shown as verified only when three different devices agree, and reports are anonymous.
- **It tells you what to do, not just what is wrong.** It learns your street's pattern, then says what to do now: charge phones, cook while there is power, fill buckets.
- **It protects the people most at risk.** The medical battery countdown, staged alerts and one-tap ambulance call exist because losing power can cost a life. Most outage tools stop at the alert.
- **It supports the local economy.** Spazas see what an outage costs them and order stock together for lower prices.
- **It is built for the conditions it serves.** It works offline, is anonymous, multilingual, light on data, and usable by people who cannot see or hear a screen alert.

5. Unfair advantage

- **No dependence on anyone else's data.** Loadshedding apps stop working when there is no official schedule. Our source is the neighbours, so the service keeps working where official systems are silent.
- **It still works when the power and signal go.** The app works offline and queues posts until signal returns.
- **The data improves with every report and stays local.** Each verified outage adds to a record of when and how long outages happen in that community. A newcomer can copy our screens but not that history or the trust around it.
- **We are close to the problem.** Two 22-year-old South Africans who live with these outages: a developer who ships, and a biomedical engineering student who knows how people depend on power for health.

6. Customer segments

Segment	Who exactly	Why them
Township households (primary)	Adults 18 to 55 in low-income households (roughly under R10,000 a month) in Gauteng to start, using prepaid electricity and often one shared smartphone on prepaid data.	They feel unscheduled outages first and cannot absorb spoiled food. They need a free, low-data, offline tool in their own language.
Spaza shops and small traders (primary)	Owner-operators aged 25 to 55 with a fridge or freezer and mobile money.	An outage means spoiled stock and lost sales the same day, and buying alone costs more.
Medically dependent households (priority)	Families using oxygen concentrators, nebulisers, CPAP or dialysis at home, and their carers (often an adult child aged 25 to 60).	The stakes are highest: they need to know how long a battery lasts and who to call.
Neighbours who can help (secondary)	People with a generator, solar kit, borehole or car; ward and street committees.	They turn the board into real help and spread the app by trust.
Partners (later)	Equipment suppliers, carers, clinics, wholesalers.	They fulfil requests and give a way to earn without charging households.

Entry point: a few suburbs in Johannesburg first, in English plus the main local languages, then other cities. The first users to reach are spaza owners and community groups, because one shop owner or street

committee brings in many households.

7. Key metrics

Targets for a first six-month Johannesburg pilot. They are goals, not results.

Area	Metric and target
Reach	3,000 active users in 10 or more suburbs; 150 spaza shops, 40 in a shared restock order; 100 medically dependent households with a battery plan
Trust and accuracy	85% of verified outages later confirmed as real; median time from first report to verified alert under 20 minutes; 60% of "likely outage" warnings followed by an outage in the stated window
Impact	50% of users who get an early warning take action; spaza owners report spoiled stock down 20%; pooled orders save at least 8%; medical households knew their battery time in 80% of outages
Engagement	40% weekly return; 30% of users have filed a report
Sustainability	At least 3 partners signed; cost per active user under R5 a month

How we measure: built-in analytics that identify no one, a short in-app survey after an outage, and follow-up calls with a sample of spaza owners and medical households.

8. Distribution channels

Kasi Grid is a web app that installs from the browser, with no app store, no download fee and no large download. Users open a link, tap Add to home screen, and it works offline.

- **WhatsApp.** Every verified outage and restock order has a one-tap share button; spaza owners invite other spazas.
- **Spaza shops as local hubs.** A QR code and poster at the counter; the owner has their own reason to promote it (restock savings).
- **Community leaders and groups.** Ward and street committees, community WhatsApp and Facebook groups, churches and stokvels.
- **Carers, clinics and equipment suppliers.** For medically dependent households, the link is handed out with the equipment or at a visit.
- **Community radio** in local languages for people without smartphones or data.
- **Pilot launches in a few suburbs** first, then repeat the approach suburb by suburb.

9. Cost structure

Estimates for a six-month Johannesburg pilot with about 3,000 users. Planning figures, not quotes.

Monthly running cost	Per month
Hosting and database (serverless host, hosted database, backups)	R600
Outage and weather data (EskomSePush API, weather feed)	R500
Domain, monitoring and error tracking	R300

Monthly running cost	Per month
Community moderators (3 people, R1,500 stipend each)	R4,500
Outreach and data (airtime for community champions, printing)	R3,000
Contingency (10%)	R890
Total	about R9,800

One-off cost	Amount
Translation review into local languages	R6,000
Carer and supplier vetting, through a partner	R5,000
POPIA and terms review	R8,000
Launch posters, QR codes and counter stands	R4,000
Total	R23,000

Six-month total: about R81,800. Cost per user: about R3.30 a month at 3,000 users. Payment fees (about 3% of each booking or restock order) are variable and are paid from the booking fee or supplier commission, so households do not carry them. Founders' time is not costed.

10. Revenue streams

Households pay nothing. Reporting, alerts, forecasts and the medical battery plan stay free, because charging would exclude the people who need it most. The money comes from businesses and partners that gain from the community the app builds.

Stream	Price	Month-6 estimate
Spaza Pro: advanced outage-cost reports and stock alerts	R49 a month	40 paying shops = R1,960
Restock commission: the wholesaler pays a share of pooled orders	4% of order value	40 shops x 2 orders x R2,500 = R200,000, so R8,000
Booking commission: oxygen, battery backups and carers	10% of the booking (about R45 on R450)	60 bookings = R2,700
Partner listing: verified carers, suppliers and clinics	R299 a month	5 partners = R1,495
Sponsorship or municipal dashboard	R3,000 a month	1 sponsor = R3,000
Total		about R17,155 a month

Against running costs of about R9,800 a month, that leaves about R7,000 a month by month 6, enough to cover the one-off costs within about three more months. These are projections that depend on the user targets above; the wholesaler commission and sponsor figure are not yet agreed with anyone.

Source code

The code is not attached to this entry. It is in a public Git repository: https://github.com/Unsolvedstar/Kasi-Grid. The README explains how to run it, and the repository holds the 111 automated tests.